

Lesson 3: Language of Probability

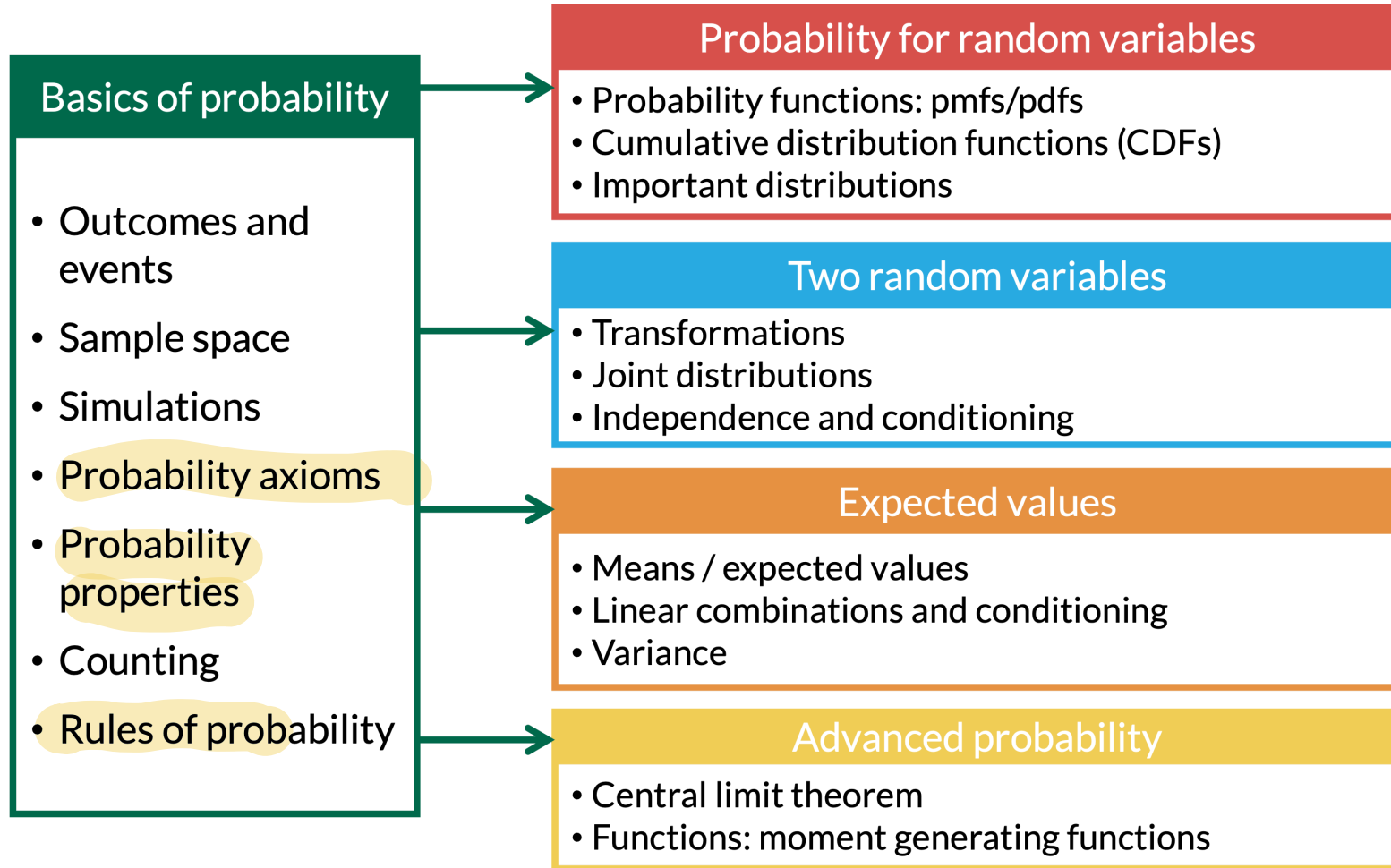
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2025-10-06

Learning Objectives

1. Use set notation, Venn diagrams, and the concepts of unions, intersections, complements, and mutually exclusive events to represent and describe events.
2. Apply the axioms of probability and related properties to calculate probabilities and prove simple results.
3. Explain and use De Morgan's Laws to simplify and solve probability problems.
4. Connect partitions and all rules of probability to calculate probabilities.

Where are we?



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Set Theory (1/2)

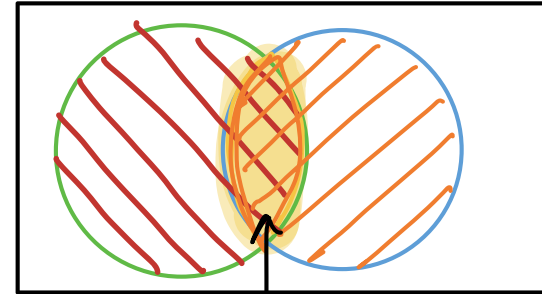
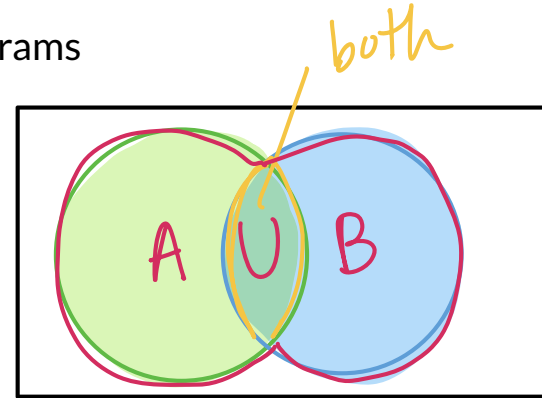
Definition: Union

The **union** of events A and B , denoted by $A \cup B$, contains all outcomes that are in A or B or both

Definition: Intersection

The **intersection** of events A and B , denoted by $A \cap B$, contains all outcomes that are both in A and B .

Venn diagrams



$A \cap B$

Set Theory (2/2)

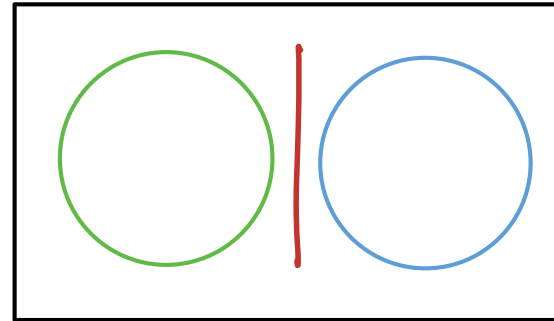
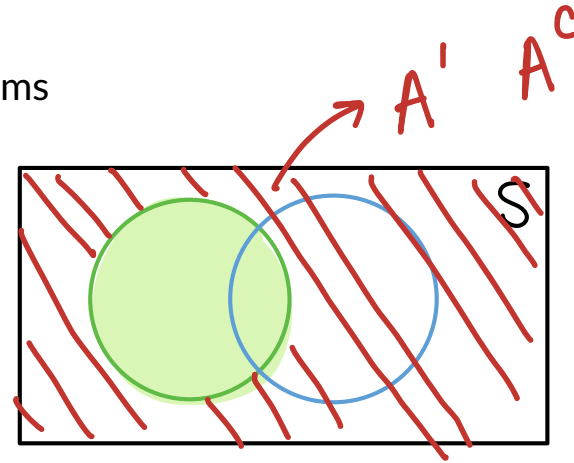
Venn diagrams

Definition: Complement

The **complement** of event A , denoted by A^C or A' , contains all outcomes in the sample space S that are **not in A** .

Definition: Mutually Exclusive

Events A and B are **mutually exclusive**, or disjoint, if they have no outcomes in common. In this case $A \cap B = \emptyset$, where $\emptyset$ is the empty set.



$$A \cap B = \emptyset$$

How can we code some of these? (1/2)

Example: Simulating Two Rolls of a Fair Four-Sided Die

We're going to roll two four-sided dice. This time, let's say event A is rolling matching numbers and event B is rolling at least one 2.

- First, we simulate rolling two four-sided dice 10,000 times

```
1 set.seed(1002) ←  
2 rolls = replicate(10000, sample(x = 1:4, size = 2, replace = TRUE))
```

- Now, we can create logical vectors for events A and B

```
1 event_A = ( rolls[1, ] == rolls[2, ] )  
2 head(event_A, 10)  
[1] FALSE FALSE FALSE FALSE FALSE FALSE TRUE FALSE FALSE FALSE
```

```
1 event_B = ( rolls[1, ] == 2 | rolls[2, ] == 2 )  
2 head(event_B, 10)  
[1] FALSE FALSE FALSE TRUE FALSE TRUE FALSE TRUE FALSE FALSE
```

Reps

	1	2	3...	10000
roll 1	1	2	4	
roll 2	1	3	2	
	T	F	F	

$B = C \cup D$
↓ ↓
C first D second
dice rolls 2 dice rolls 2

How can we code some of these? (2/2)

Union

$$A \cup B$$

$$A \mid B$$

```
1 event_A_or_B = event_A | event_B
2 head(event_A_or_B, 10)
```

```
[1] FALSE FALSE FALSE TRUE FALSE TRUE TRUE
TRUE FALSE FALSE
```

Intersection

$$A \cap B \quad A \& B$$

```
1 event_A_and_B = event_A & event_B
2 head(event_A_and_B, 10)
```

```
[1] FALSE FALSE FALSE FALSE FALSE FALSE FALSE
FALSE FALSE FALSE
```

Complement

$$A^c \text{ or } A' \neq A$$

```
1 event_not_A = !event_A
2 event_not_B = event_B != TRUE
3 head(event_not_A, 10)
```

```
[1] TRUE TRUE TRUE TRUE TRUE TRUE TRUE FALSE
TRUE TRUE TRUE
```

Mutually Exclusive

$$A \cap B = \emptyset \quad A \& B == NA$$

```
1 sum(event_A_and_B == TRUE)
```

```
[1] 621
```

↳ 0

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Probability Axioms

Axiom 1

For every event A , $0 \leq \mathbb{P}(A) \leq 1$. Probability is between 0 and 1.

Axiom 2

For the sample space S , $\mathbb{P}(S) = 1$.

Axiom 3

If $A_1, A_2, A_3, \dots$, is a collection of **disjoint** events, then

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i).$$

The probability of at least one A_i is the sum of the individual probabilities of each.

Chapter 2

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$$P(A) = \frac{|A|}{|S|}$$

$$P(S) = \frac{|S|}{|S|} = 1$$

$$\begin{aligned} P(A_1 \cup A_2 \cup A_3 \cup \dots \cup A_{\infty}) \\ = P(A_1) + P(A_2) + P(A_3) \\ + \dots + P(A_{\infty}) \end{aligned}$$

$$P(A \cup B) = P(A) + P(B)$$

only if disjoint/mutually exc.

Some probability properties

Using the Axioms, we can prove all other probability properties! Events A , B , and C are not necessarily disjoint!

Proposition 1

For any event A , $\mathbb{P}(A) = 1 - \mathbb{P}(A^C)$

Proposition 2

$\mathbb{P}(\emptyset) = 0$

Proposition 3

If $A \subseteq B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$

Proposition 4

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$$

where A and B are not necessarily disjoint



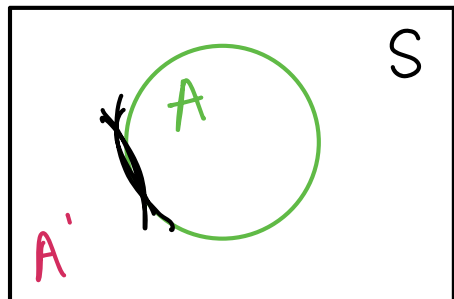
Proposition 5

$$\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) - \mathbb{P}(B \cap C) + \mathbb{P}(A \cap B \cap C)$$

Proposition 1 Proof

Proposition 1

For any event A , $\mathbb{P}(A) = 1 - \mathbb{P}(A^C)$



A & A' are
disjoint

$$A \cup A' = S$$

$$\mathbb{P}(A \cup A') = \mathbb{P}(S) \\ \hookrightarrow = 1$$

$$\mathbb{P}(A \cup A') = 1 \\ \downarrow$$

$$\mathbb{P}(A) + \mathbb{P}(A') = 1 \\ -\mathbb{P}(A') \quad -\mathbb{P}(A') \\ \mathbb{P}(A) = 1 - \mathbb{P}(A')$$

Use Axioms!

A1: $0 \leq \mathbb{P}(A) \leq 1$

A2: $\mathbb{P}(S) = 1$

A3: For disjoint A_i ,

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$$

Proposition 2 Proof

Proposition 2

$$\mathbb{P}(\emptyset) = 0$$

$$\text{prop 1: } P(A) = 1 - P(A^c)$$

$$\text{let } A = \emptyset \quad A^c = S$$

$$P(\emptyset) = 1 - P(S)$$
$$\hookrightarrow = 1$$

$$P(\emptyset) = 1 - 1 = 0$$

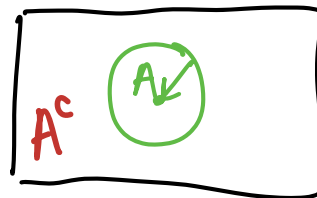
Use Axioms!

$$\text{A1: } 0 \leq \mathbb{P}(A) \leq 1$$

$$\text{A2: } \mathbb{P}(S) = 1$$

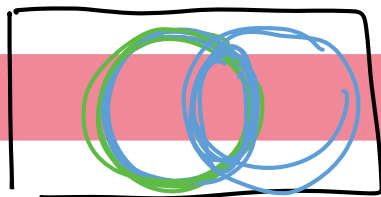
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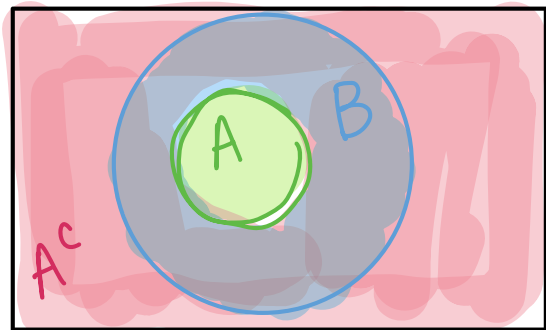
Proposition 3 Proof

$$\underline{P(A)} \quad \text{vs} \quad \underline{P(A \cap B)}$$



Proposition 3

If $A \subseteq B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$



$$B = \underbrace{A} \cup \underbrace{(B \cap A^c)}$$

b/c disjoint

$$\underline{P(B)} = \underline{P(A)} + \underline{P(B \cap A^c)}$$

A & A^c
are disjoint
 $\underline{(B \cap A^c)} \subseteq A^c$

$$P(B) \geq P(A)$$

Use Axioms!

$$A1: 0 \leq \mathbb{P}(A) \leq 1$$

$$A2: \mathbb{P}(S) = 1$$

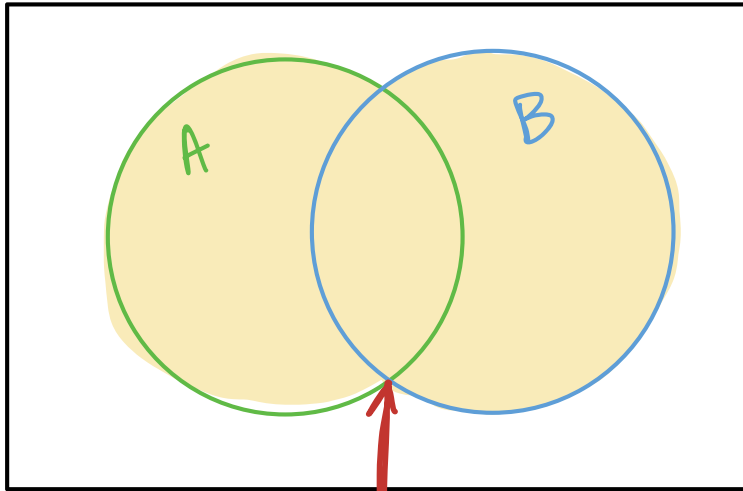
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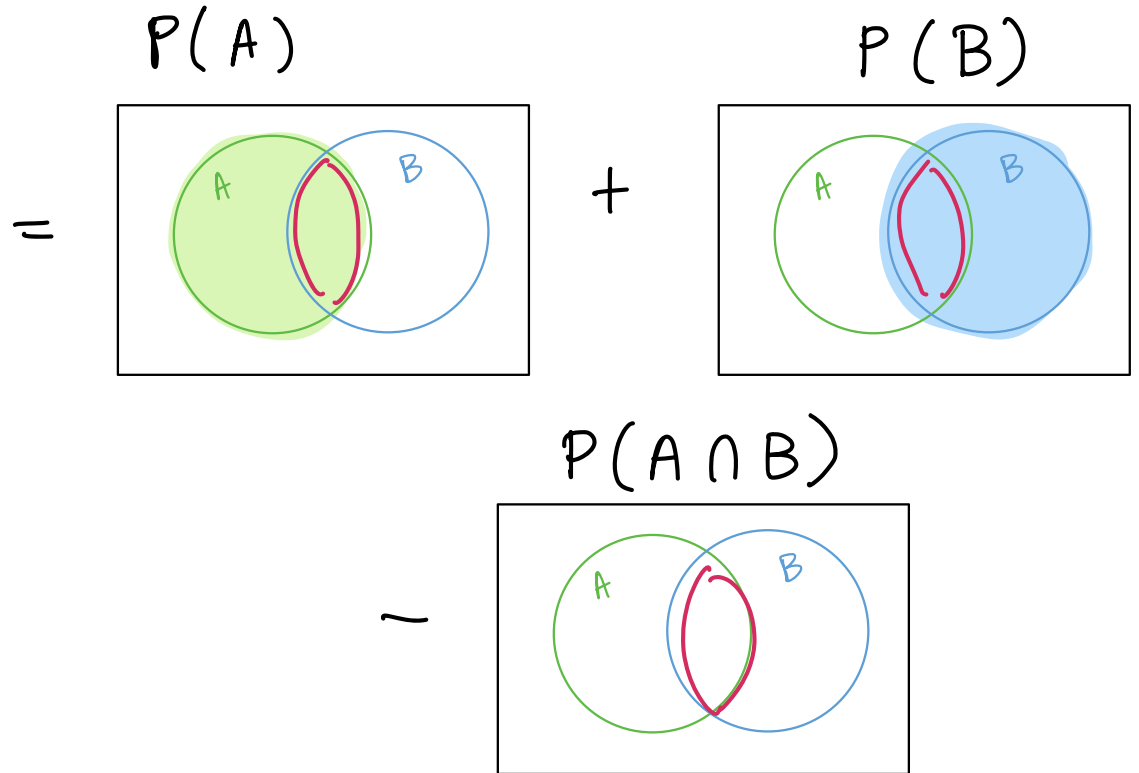
Proposition 4 Visual Proof

Proposition 4

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$$



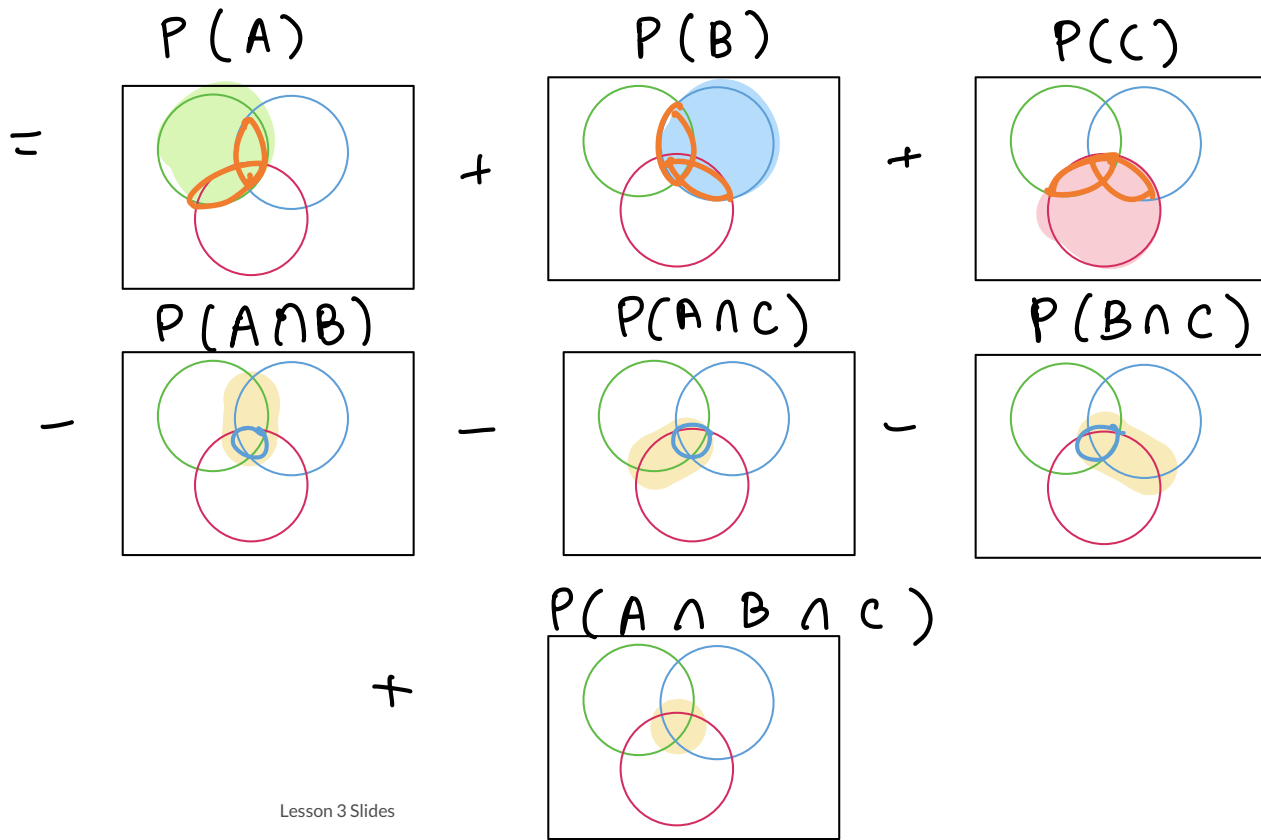
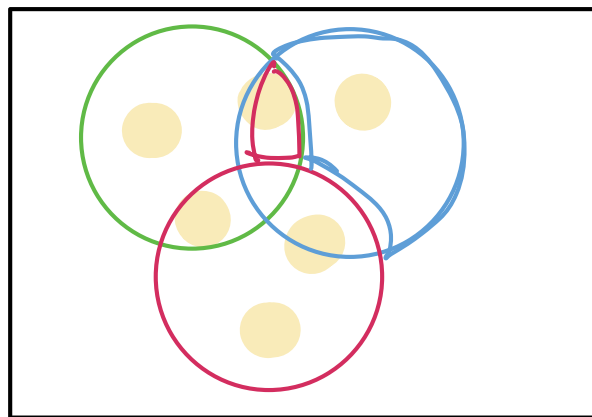
$$\mathbb{P}(A \cap B) \geq 0$$



Proposition 5 Visual Proof

Proposition 5

$$\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) - \mathbb{P}(B \cap C) + \mathbb{P}(A \cap B \cap C)$$



Some final remarks on these proposition

- Notice how we spliced events into multiple **disjoint** events
 - It is often easier to work with disjoint events
- If we want to calculate the probability for one event, we may need to get creative with how we manipulate other events and the sample space
 - Helps us use any incomplete information we have

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De Morgan's Laws

$$\sum_{i=1}^n P(A_i) = P(A_1) + P(A_2) + \dots + P(A_n)$$

Theorem: De Morgan's 1st Law

For a collection of events (sets) $A_1, A_2, A_3, \dots$

$$\bigcap_{i=1}^n A_i^C = \left(\bigcup_{i=1}^n A_i \right)^C$$

$$\bigcap_{i=1}^n B_i = B_1 \cap B_2 \cap B_3 \cap \dots \cap B_n$$

“all not A = (at least one event A)^C” or “intersection of the complements is the complement of the union”

Theorem: De Morgan's 2nd Law

For a collection of events (sets) $A_1, A_2, A_3, \dots$

$$\bigcup_{i=1}^n A_i^C = \left(\bigcap_{i=1}^n A_i \right)^C$$

“at least one event not A = (all A)^C” or “union of complements is complement of the intersection”

BP example variation (1/3)

- Suppose you have n subjects in a study.
- Let H_i be the event that person i has high BP, for $i = 1 \dots n$.

Use set theory notation to denote the following events:

1. Event subject i does not have high BP
2. Event all n subjects have high BP
3. Event at least one subject has high BP
4. Event all of them do not have high BP
5. Event at least one subject does not have high BP

BP example variation (2/3)

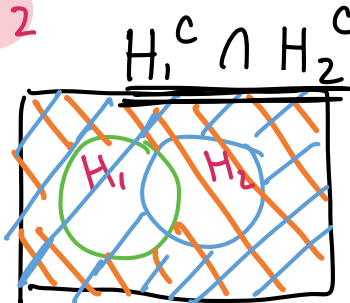
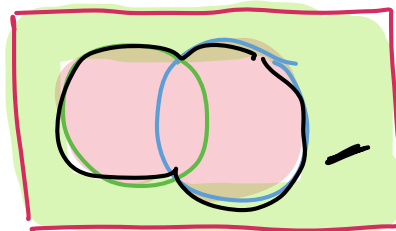
- Suppose you have n subjects in a study.
- Let H_i be the event that person i has high BP, for $i = 1 \dots n$.

Use set theory notation to denote the following events:

1. Event subject i does not have high BP H_i' or H_i^c

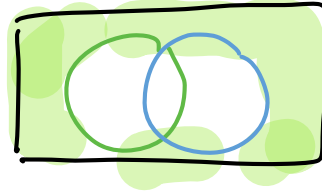
2. Event all n subjects have high BP
 $P1 \ \& \ P2: H_1 \cap H_2 \xrightarrow{to} H_1 \cap H_2 \cap H_3 \cap \dots \cap H_n = \bigcap_{i=1}^n H_i$

3. Event at least one subject has high BP
 $P1 \ \& \ P2: H_1 \cup H_2 \rightarrow \underline{H_1 \cup H_2} = \left(H_1^c \cap H_2^c \right)^c$
 $\bigcup_{i=1}^n H_i = \left(\bigcap_{i=1}^n H_i^c \right)^c$



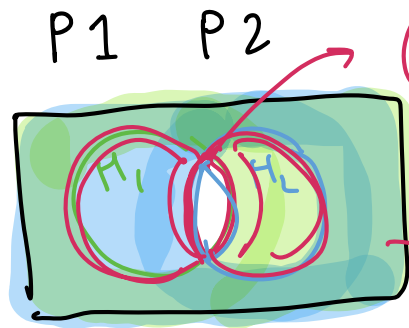
BP example variation (3/3)

4. Event all of them do not have high BP



$$\bigcap_{i=1}^n H_i^c$$

5. Event at least one subject does not have high BP



$$(H_1 \cap H_2)^c$$

$$(H_1 \cap H_2 \cap H_3 \cap \dots \cap H_n)^c$$

$$H_1^c \cup H_2^c$$

$$= \left(\bigcap_{i=1}^n H_i \right)^c$$

everything but
Both having high BP

$$= \left(\bigcup_{i=1}^n H_i^c \right)$$

Remarks on De Morgan's Laws

- These laws also hold for infinite collections of events.
- Draw Venn diagrams to convince yourself that these are true!
- These laws are very useful when calculating probabilities.
 - This is because calculating the probability of the **intersection of events is often much easier than the union of events**.
 - This is not obvious right now, but we will see in the coming chapters why.

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Partitions

Definition: Partition

A set of events $\{A_i\}_{i=1}^n$ create a **partition** of A , if

- the A_i 's are disjoint (mutually exclusive) and
- $\bigcup_{i=1}^n A_i = A$

Example 2

- If $A \subset B$, then $\{A, B \cap A^C\}$ is a partition of B .
- If $S = \bigcup_{i=1}^n A_i$, and the A_i 's are disjoint, then the A_i 's are a partition of the sample space.

Creating partitions is sometimes used to help calculate probabilities, since by Axiom 3 we can add the probabilities of disjoint events.

Weekly medications

Example 3

If a subject has an

- 80% chance of taking their medication *this* week,
- 70% chance of taking their medication *next* week, and
- 10% chance of *not* taking their medication *either* week,

then find the probability of them taking their medication exactly one of the two weeks.

Hint: Draw a Venn diagram labelling each of the parts to find the probability.

